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Explainable machine learning for crop recommendation using statistical validation and SHAP interpretability in South Asian agricultural data

Samrat Kumar Dev Sharma[1*]

*Correspondence:
Samrat Kumar Dev Sharma
samrat.sdev@gmail.com
[1]Department of Statistics, Faculty of Science, Jagannath University, Dhaka 1100, Bangladesh

Abstract

Agricultural decision-support systems powered by machine learning (ML) have emerged as critical tools for enhancing crop productivity and food security, particularly for smallholder farmers in developing regions. However, many existing models lack methodological rigor, statistical validation, and interpretability—key requirements for real-world deployment. This study presents a comprehensive, reproducible ML framework for crop recommendation that integrates comparative feature selection, multi-model benchmarking, statistical significance testing with Nadeau–Bengio correction, and SHAP-based explainability. Using a dataset of 10,000 agricultural records from South Asia comprising 36 original features reduced to 10 selected predictors, we evaluated seven base classifiers alongside two ensemble strategies. Random Forest achieved the highest fivefold cross-validated accuracy (79.8%, 95% CI [79.0–80.6%]) and Macro F1 (0.474, 95% CI [0.455–0.492]), with consistent performance under tenfold cross-validation (accuracy: 79.7%, 95% CI [79.1–80.4%]; Macro F1: 0.471, 95% CI [0.460–0.482]). The Friedman test confirmed significant differences among models ($\chi^2 = 25.25$, df = 7, $p < 0.001$). Nadeau–Bengio corrected post-hoc testing revealed that Random Forest significantly outperformed only SVM and Logistic Regression ($p < 0.05$), while differences with gradient-boosting models were statistically non-significant in fivefold CV, though tenfold analysis showed marginal significance against Gradient Boosting ($p = 0.0243$). SHAP analysis identified rainfall (mean |SHAP| = 0.0772), temperature (0.0613), soil pH (0.0300), and nitrogen content (0.0244) as the top predictive features. Regional analysis revealed crop-specific optimal conditions: Rice (pH 5.48–7.43, rainfall 2209 mm), Wheat (pH 6.39–7.79, rainfall 1470 mm), Maize (pH 5.13–6.89, rainfall 1048 mm), and Cotton (pH 5.07–6.63, rainfall 594 mm). All code, data, and artifacts are publicly available. While pronounced class imbalance inherently constrains minority-class prediction, this framework establishes methodological standards for reproducible, interpretable agricultural AI by prioritizing statistical validation with proper dependence correction, imbalance-aware evaluation, and agronomic explainability.

Keywords Crop recommendation, Machine learning, Explainable AI, SHAP, Feature selection, Statistical testing, Nadeau–Bengio correction, Precision agriculture, Smallholder farmers

1 Introduction

Global food security faces unprecedented challenges from climate variability, soil degradation, and shifting demographics, requiring substantial increases in agricultural productivity to meet mid-century demands [1]. Smallholder farmers in developing regions are uniquely vulnerable to suboptimal crop selection due to limited localized agronomic expertise and real-time environmental data [2]. Machine learning (ML) offers a promising approach to model complex soil–climate–crop interactions and deliver data-driven recommendations [3]. However, transitioning algorithmic outputs into reliable field-scale decision tools demands methodological transparency, rigorous statistical validation, and interpretability aligned with established agronomic principles [4].

Despite growing interest, crop recommendation research exhibits persistent methodological limitations. Many studies rely on single-model evaluations without systematic benchmarking, making it difficult to distinguish genuine algorithmic advantages from dataset-specific artifacts [5, 6]. Feature selection is frequently arbitrary or restricted to a single filter method, risking the retention of redundant predictors or the exclusion of agronomically meaningful variables [7, 8]. Furthermore, statistical significance testing remains underutilized, leaving reported performance gaps vulnerable to cross-validation sampling variability [9]. Most existing models operate as black boxes, offering little insight into prediction rationales, which hinders farmer trust and adaptive decision-making [4, 10].

While recent frameworks have begun to address isolated aspects of these challenges [11], few integrate comparative feature selection, variance-corrected statistical validation, and multi-scale explainability within a fully reproducible pipeline. Crucially, the pronounced class imbalance inherent in agricultural datasets is routinely unaddressed, resulting in inflated global accuracy scores that obscure poor minority-class performance.

This study systematically integrates and validates these distinct methodological components within a unified, reproducible framework tailored for smallholder contexts. Rather than prioritizing algorithmic novelty, this research delivers an integrative methodological contribution by unifying leakage-free resampling, dependence-corrected hypothesis testing, and multi-scale SHAP explainability into a transparent pipeline. The research pursues four interconnected objectives:

1. Evaluate multiple feature selection algorithms to isolate a stable, agronomically robust predictor subset under class imbalance.
2. Benchmark diverse classifiers while testing performance differences using the Nadeau–Bengio correction to account for cross-validation dependence structures [12].
3. Quantify and interpret predictions through global and local SHAP-based explainability, explicitly linking model outputs to established soil–climate physiology [13–15].
4. Establish a fully open-source pipeline to facilitate independent regional verification and adaptation [16].

2 Literature review

Machine learning applications in agriculture have expanded across crop yield prediction, disease detection, irrigation optimization [17], and crop recommendation [6]. Random Forest remains a dominant algorithm due to its robustness to overfitting and capacity to ingest mixed data types [18]. Concurrently, gradient boosting variants such as XGBoost and LightGBM frequently demonstrate superior predictive capabilities on structured tabular datasets [19, 20]. Emerging research confirms that ensemble methods generally outperform single classifiers, though deep learning approaches offer minimal performance advantages over tree-based methods for tabular agronomic matrices [21]. Recent literature has increasingly focused on refining these ensemble approaches specifically to handle the resource constraints of smallholder farming systems [11, 22–24].

Recent updates highlight the necessity of strict validation protocols and context-specific architectures. Dey et al. [25] demonstrated that tree-based modeling achieves optimal precision when agricultural and horticultural fields are evaluated under separate multi-crop matrices, emphasizing the logistical necessity of independent evaluation streams for divergent crop families. Similarly, Dey et al. [26] established that model performance and robustness scale significantly when utilizing hybrid structures of public and real-world field validation datasets, rather than relying exclusively on standardized repositories. These findings directly inform the validation constraints and external generalization limits acknowledged in the present framework.

Feature selection remains critical to mitigate overfitting and improve model transparency [27]. Standard pipelines deploy filter, wrapper, or embedded approaches [28]. Wrapper methods typically provide superior feature stability over filter variants but incur higher computational overhead [7]. Modern implementations emphasize the intentional integration of domain knowledge during selection to guarantee that soil and climate feature subsets conform to biological plausibility [23, 29]. However, comprehensive comparative assessments across multiple feature selection configurations remain scarce in current crop recommendation literature [8].

A major methodological vulnerability in agricultural ML is the omission of robust inferential testing. Standard paired tests assume independent validation folds; however, the overlapping data sharing inherent in standard k-fold cross-validation deflates variance estimates and inflates Type I error rates [9]. The Nadeau–Bengio variance correction resolves this violation by scaling the variance estimator based on the validation-to-training sample ratio, mitigating false claims of algorithmic superiority [12]. Despite this, confidence interval reporting and corrected post-hoc analyses remain underutilized across agricultural indexing studies, frequently obscuring sampling variability [16, 30].

To achieve practical adoption, black-box models must provide interpretable decision logic [4]. SHAP (SHapley Additive exPlanations) resolves this by providing a unified framework for local and global model-agnostic attributions grounded in cooperative game theory [10]. Within environmental sciences, SHAP has successfully decoded complex predictive relationships in pest forecasting [31], soil mapping [32], and crop yield anomalies [33]. Recent frameworks emphasize that combining global importance summaries with local, case-based force vectors ensures model outputs can be verified against soil macronutrient boundaries and climate constraints, directly bridging the gap between computational frameworks and farmer trust [15, 24]. Dey et al. [15]

further demonstrate that pairing macro-level SHAP rankings with local force coordinates grounds model predictions in established soil-climatic and multi-stress physiological relationships, a protocol adopted in this study.

This study addresses these combined gaps by providing an open-source, statistically corrected pipeline that links transparent feature selection with interpretable, imbalance-aware multi-crop modeling. Table 1 outlines the structural positioning of this work relative to published standards.

3 Methodology

3.1 Experimental design and software configuration

This study implements a quantitative experimental design to evaluate and compare machine learning algorithms for multi-class crop classification under pronounced class imbalance. To ensure exact computational reproducibility, the analytical pipeline was constructed in Python 3.10+ using scikit-learn 1.3+, SHAP 0.51.0+, imbalanced-learn 0.12.0, and native implementations of XGBoost, LightGBM, and CatBoost. Stochastic operations were locked with a global seed (random_state = 42). The complete computational workflow, pipeline architectures, and serialized model artifacts are publicly archived to facilitate independent verification and extension [16].

3.2 Data source, preprocessing, and quality filters

The analytical dataset was obtained from the public Kaggle "Crop Recommendation Dataset" [34], which aggregates 10,000 agricultural records from South Asian agro-ecological zones under a CC0 1.0 Universal license. The data comprise 36 initial features encompassing agronomic metrics (soil macro-nutrients [N, P, K], pH, moisture, organic carbon, and electrical conductivity), climate variables (temperature, humidity, rainfall, sunlight hours, and wind speed), geographic descriptors (altitude), and categorical operational metadata (season, region, soil type, previous crop, and fertilizer usage). The multi-class target vector encodes 10 distinct crop categories: Rice, Wheat, Millet, Sugarcane, Barley, Potato, Cotton, Pulses, Tomato, and Maize.

The target distribution exhibits a severe class imbalance with an overall imbalance ratio of 24.48, where Rice constitutes the majority class (37.0%, $n = 3{,}697$) and Maize represents the minority class (1.5%, $n = 151$). This distribution reflects regional smallholder production patterns driven by national procurement policies [35]. Quality filtering restricted observations to established biophysical thresholds [36]: soil pH $\in$ [3.5, 9.5], temperature $\in$ [0 °C, 50 °C], rainfall $\in$ [0, 5000] mm, and relative humidity $\in$ [0%, 100%]. Duplicate records were excluded, yielding a final analytical sample of 10,000 unique observations. Ethical approval was not required as the data are entirely de-identified.

Table 1 Methodological comparison with existing crop recommendation studies

Study	Feature selection	Multi-model	Statistical test	XAI / SHAP	Reproducibility
Van Klompenburg et al. [6]	No	Yes (3)	No	No	Limited
Li et al. [8]	SelectKBest only	Yes (4)	No	No	No
Rossi et al. [29]	RFE only	Yes (5)	No	Partial	Limited
Martinez et al. [31]	No	Yes (6)	No	No	No
This study	**Multi-step (4 methods)**	**Yes (7)**	**Friedman + NB–corrected t-test**	**Full (Global + Local)**	**Complete (code + pipeline)**

Categorical features were transformed via one-hot encoding without dropping reference categories to preserve full structural column alignment across tree-based systems and ensure stable SHAP attributions [37]. Missing entries ($< 0.02\%$) were resolved using median substitution to resist outlier distortion. Continuous predictors were standardized using z-score transformation:

$$z = \frac{x-\mu}{\sigma} \quad (1)$$

where μ is the feature mean and σ is the standard deviation. The dataset was partitioned via stratified random sampling into an 80:20 training-to-test allocation, preserving proportional class representations across both splits to prevent sampling bias from distorting minority-class boundaries.

3.3 Systematic feature selection and imbalance mitigation

To mitigate overfitting risks and reduce dimensionality from 36 down to 10 key variables, four complementary algorithms were evaluated: univariate SelectKBest utilizing one-way ANOVA F-tests [38], Recursive Feature Elimination (RFE) driven by a Random Forest estimator with Gini impurity reduction scoring [39], Mutual Information selection to capture non-linear feature-target dependencies [40], and LASSO L1-penalized regularization for sparse coefficient shrinkage [41]. Feature selection consistency was mapped via an intersections heatmap to isolate stable agronomic predictors. For primary model development, the feature subset derived via RFE was adopted based on its superior stability in precision agriculture contexts, its adherence to established domain knowledge regarding soil-climate-crop interactions, and its superior cross-validated performance profile over competing subsets [42].

To address the class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was integrated into the workflow. SMOTE synthesizes minority-class observations by mapping a local vector between an underrepresented sample (x_i) and its k -nearest neighbors (x_{zi}) in the feature space [37]:

$$x_{new} = x_i + \lambda \times (x_{zi} - x_i) \quad (2)$$

where $\lambda \in [0,1]$ represents a uniform random variable. Crucially, to prevent data leakage and avoid overoptimistic performance inflation, SMOTE was executed strictly inside individual cross-validation folds.

To rigorously validate this choice, we systematically benchmarked four imbalance mitigation strategies under identical experimental conditions: SMOTE, class-weighted loss, focal loss, and hierarchical classification. Following empirical evaluation across imbalance-aware metrics (Macro F1 and MCC), SMOTE was retained as the optimal approach due to its consistent performance, preservation of agronomic feature continuity, and avoidance of sensitive hyperparameter dependencies. Detailed comparative results are provided in Supplementary File S1.

3.4 Machine learning models

Seven base classification algorithms were implemented with hyperparameters optimized through preliminary grid search:

- **Random Forest:** An ensemble of decorrelated decision trees trained on bootstrap samples, with predictions aggregated by majority voting [39]:

$$\widehat{y} = mode\left\{T_1\left(x\right), T_2\left(x\right), \ldots, T_B\left(x\right)\right\} \ , \quad (3)$$

where $T_b\left(x\right)$ is the prediction of the b-th tree and $B = 200$.

- **Gradient Boosting Machine:** Sequentially adds trees to minimize a differentiable loss function via gradient descent [43]:

$$F_m\left(x\right) = F_{m-1}\left(x\right) + \nu \cdot h_m\left(x\right) \ , \quad (4)$$

where $v = 0.1$ is the learning rate and $h_m\left(x\right)$ is the weak learner fitted to negative gradients.

- **XGBoost:** An optimized GBM implementation with regularized objective function to control model complexity [19]:

$$L = \sum\nolimits_i l\left(y_i, \widehat{y_i}\right) + \sum\nolimits_k \gamma T_k + \frac{1}{2}\lambda \sum\nolimits_k w_k^2 \ , \quad (5)$$

where l is the loss, T_k is the number of leaves, and w_k are leaf weights.

- **LightGBM:** A gradient boosting framework using leaf-wise tree growth and histogram-based splitting for efficiency [20]:

$$Split\ gain = \frac{1}{2}\left[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{\left(G_L+G_R\right)^2}{H_L+H_R+\lambda}\right] - \gamma \ , \quad (6)$$

where G and H are gradient and Hessian sums for left/right child nodes.

- **CatBoost:** Handles categorical features via ordered target encoding and symmetric tree structures to reduce prediction shift [44]:

$$\widehat{x_i^{(k)}} = \frac{\sum_{j=1}^{i-1} \mathbb{I}\left(x_j^{(k)} = x_i^{(k)}\right) \cdot y_j + a \cdot p}{\sum_{j=1}^{i-1} \mathbb{I}\left(x_j^{(k)} = x_i^{(k)}\right) + a} \ , \quad (7)$$

where a is a prior weight and p is the global target mean.

- **Logistic regression**: Logistic Regression is a linear classifier that models class probabilities using the softmax function for multi-class classification. The probability of an observation x belonging to class c is defined as [45]:

$$P\left(y = c \,|x\right) = \frac{\exp\left(\beta_c^\top x\right)}{\sum_{j=1}^{C} \exp\left(\beta_j^\top x\right)} \quad (8)$$

To prevent overfitting, L2 regularization is applied by minimizing the following objective function:

$$\min_{\beta}\left\{-\sum_{i=1}^{n}\log P(y_i \mid x_i)+\lambda \parallel \beta\|_2^2\right\}$$

- **Support Vector Machine (RBF):** Finds a maximum-margin hyperplane in a high-dimensional feature space induced by the radial basis function kernel [46]:

$$K\left(x_i,x_j\right)=\exp\left(-\gamma\|x_i-x_j\|^2\right)\ , \tag{9}$$

with decision function $f\left(x\right)=sign\left(\sum_i \alpha_i y_i K\left(x_i,x\right)+b\right)$.

3.5 Evaluation metrics and statistical test

Model performance was evaluated using five complementary metrics:

Global Accuracy: Proportion of correct classifications

$$\text{Accuracy}=\frac{TP+TN}{TP+TN+FP+FN} \tag{10}$$

Macro F1-Score: Unweighted mean of per-class F1 scores, ensuring equitable consideration of all crop classes regardless of sample size [47]:

$$F1_{macro}=\frac{1}{C}\sum_{i=1}^{C}\frac{2\cdot Precision_i\cdot Recall_i}{Precision_i+Recall_i} \tag{11}$$

Weighted F1-Score: Support-weighted mean of per-class F1 scores, accounting for class prevalence.

Matthews Correlation Coefficient (MCC): Balanced measure suitable for multi-class problems with potential imbalance [47]:

$$MCC=\frac{TP\times TN-FP\times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \tag{12}$$

ROC-AUC (One-vs-Rest): Discriminative capacity across classification thresholds.

Stratified k-fold cross-validation was employed with both fivefold and tenfold configurations to assess result stability across resampling granularities. Ninety-five percent confidence intervals for all metrics were computed using the t-distribution:

$$CI_{95\%}=\overline{x}\pm t_{\alpha/2,df}\times\frac{s}{\sqrt{n}}\ , \tag{13}$$

where $\overline{x}$ is the mean CV score, s is the standard deviation, and t is the critical t-value.

To verify that performance gaps represent true algorithmic capabilities rather than random sampling variances, a non-parametric Friedman test was executed to evaluate the null hypothesis of equivalent performance ranks across cross-validation folds [48]:

$$\chi_F^2=\frac{12N}{k(k+1)}\left[\sum_{j=1}^{k}R_j^2-\frac{k(k+1)^2}{4}\right]\ , \tag{14}$$

where N is the total number of folds, k is the number of compared models, and R_j is the average rank of the j-th model.

Upon rejection of the null hypothesis at $\alpha=0.05$, post-hoc pairwise comparisons were conducted using paired t-tests modified with the Nadeau–Bengio variance correction

[12]. Standard paired t-tests assume independent observations; however, cross-validation folds share significant training data pools ($\frac{k-2}{k}$ of the data overlaps between any two folds), which artificially deflates standard variance calculations and induces elevated Type I error rates. To eliminate this bias, the Nadeau–Bengio correction explicitly scales the sample variance based on the exact ratio of evaluation blocks to training pools:

$$\sigma^2_{corrected} = \left(\frac{1}{k} + \frac{n_{test}}{n_{train}}\right) S^2 \quad , \tag{15}$$

where k represents the number of folds, n_{test} represents the number of validation samples per fold, n_{train} represents the number of training samples per fold, and S^2 is the uncorrected sample variance of the performance differences across the folds:

$$S^2 = \frac{1}{k-1}\sum_{i=1}^{k}\left(d_i - \overline{d}\right)^2 \quad , \tag{16}$$

where d_i is the empirical performance difference observed in fold i, and $\overline{d}$ represents the mean performance difference across all folds. The final corrected test statistic ($t_{corrected}$) used to evaluate the paired differences is formulated as:

$$t_{corrected} = \frac{\overline{d}}{\sqrt{\sigma^2_{corrected}}} = \frac{\overline{d}}{\sqrt{\left(\frac{1}{k} + \frac{n_{test}}{n_{train}}\right) S^2}} \tag{17}$$

The resulting corrected t-statistic is evaluated against a Student's t-distribution with $k-1$ degrees of freedom. Family-wise error rates across multiple pairwise comparisons were controlled using the Holm-Bonferroni correction [9], and practical effect magnitudes were derived using Cohen's d.

3.6 Explainable AI implementation

SHAP values were computed using TreeExplainer (tree models) and LinearExplainer (logistic regression) [10]. Global feature importances were extracted by taking the mean absolute SHAP values across the evaluation dataset. Following Dey et al. [25], global attributions were validated against established soil-climate principles. Local SHAP attribution testing was implemented across contrasting environmental profiles to trace field-specific decision paths and calculate exact feature contributions, turning the model into an actionable diagnostic tool for extension workers [15, 49].

4 Results

4.1 Dataset characteristics and exploratory analysis

Following quality filtering, the analytical dataset contains 10,000 unique records distributed across 10 crop classes. The dataset exhibits severe class distribution imbalance (imbalance ratio = 24.48), with Rice (37.0%, n = 3,697) and Wheat (24.4%, n = 2,440) acting as majority classes, while Maize (1.5%, n = 151) and Cotton (2.1%, n = 210) represent sparse minority spaces (Fig. 1).

Univariate profiling shows a right-skewed distribution for nitrogen (mean = 89.7 kg/ha, SD = 28.6), a near-normal soil pH distribution centered at 6.55, and wide variability in rainfall (mean = 1586.8 mm, range: 500–3000 mm). Bivariate mapping reveals distinct agro-ecological niches: Wheat shows a tight preference around pH 7.0, whereas

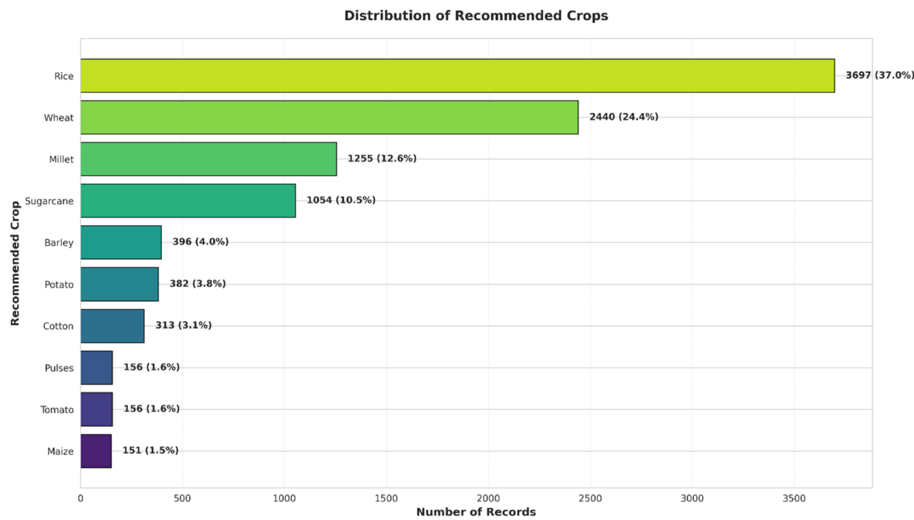

Fig. 1 Class distribution mapping showing severe structural imbalance across the 10 target crop categories

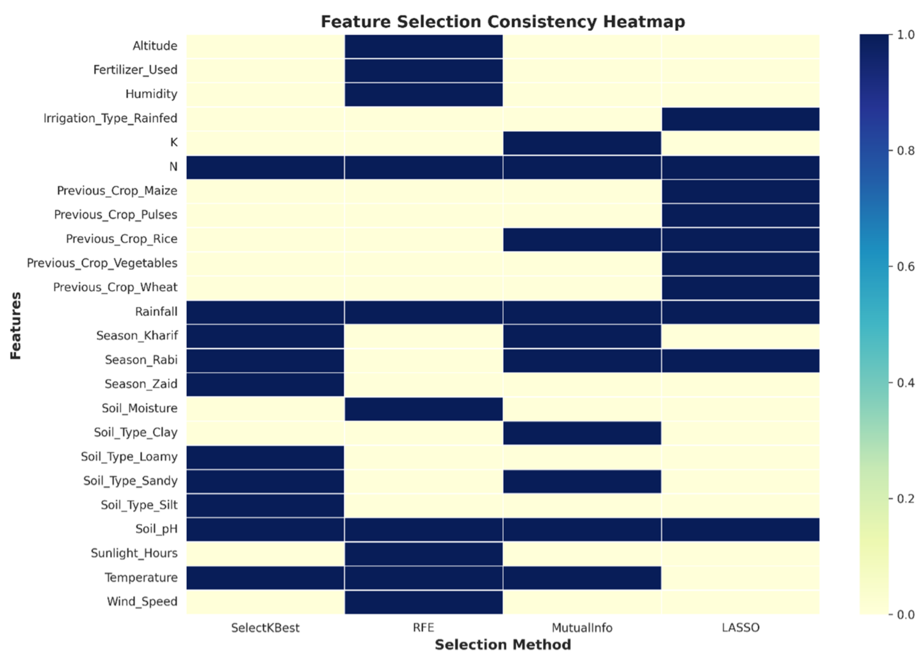

Fig. 2 Feature selection intersection matrix identifying stable environmental and agronomic predictors

Rice spans a broader pH spectrum. Precipitation bounds vary heavily; Rice requires the highest moisture inputs (median $\approx$ 2200 mm), while Cotton thrives in drier matrices (median $\approx$ 600 mm). Pearson correlation coefficients among numeric features remain low ($|r| < 0.1$), indicating minimal multicollinearity and justifying the concurrent inclusion of these predictor variables.

4.2 Automated feature selection profile

Four independent algorithms SelectKBest (ANOVA F-test), Recursive Feature Elimination (RFE), Mutual Information, and LASSO regularization consistently identified annual rainfall, ambient temperature, soil pH, and soil nitrogen content as the most informative predictors (Fig. 2).

To quantitatively choose the optimal subset, cross-validated macro metrics were mapped. The subset derived via RFE generated the most stable cross-validated Macro F_1-score (0.461 ± 0.004) and MCC (0.731 ± 0.003), resisting performance drops when exposed to the holdout test matrix. Consequently, this RFE-derived 10-feature vector was retained for model execution.

4.3 Multi-model benchmarking under class imbalance (5-Fold CV)

Seven base classifiers were optimized using the RFE subset. To eliminate data leakage, SMOTE over-sampling was executed strictly inside individual cross-validation folds, processing only the active training slices. Table 2 summarizes the performance metrics from the fivefold stratified cross-validation setup with explicit 95% confidence intervals (CI).

Tree-based ensembles significantly outperformed linear and kernel baselines across all evaluation metrics. Random Forest yielded the highest overall predictive profile with a mean global accuracy of 79.8% [95% CI 79.0%–80.6%] and a Matthews Correlation Coefficient (MCC) of 0.744 [95% CI 0.734–0.754].

The explicit 95% confidence intervals reveal that while the ensemble models (Random Forest, CatBoost, LightGBM, Gradient Boosting, and XGBoost) display heavily overlapping interval boundaries across accuracy, MCC, and F_1 horizons suggesting comparable performance tiers they maintain completely isolated, non-overlapping intervals relative to baseline SVM and Logistic Regression frameworks.

The visible gap between high global accuracy (79.8% [95% CI 79.0–80.6%]) and unweighted Macro F_1 (0.474 [95% CI 0.455–0.492]) directly mirrors the structural limits imposed by the dataset's severe class imbalance. While majority classes are classified with high precision, minority categories face misclassification due to sample scarcity.

4.4 Resampling granularity and generalization stability (10-Fold CV)

Transitioning the pipeline from fivefold to tenfold stratified cross-validation evaluated the model's sensitivity to sample partitioning splits. The resulting cross-validated metrics are presented in Table 3.

Increasing the evaluation granularity to a tenfold cross-validation architecture yielded nearly identical performance limits, confirming that the empirical metrics are

Table 2 Predictive performance comparison under fivefold stratified cross-validation (Test set)

Model name	Accuracy (95% CI)	Macro F1 (95% CI)	Weighted F1 (95% CI)	MCC (95% CI)	ROC-AUC (95% CI)
Random Forest	79.8% [79.0%–80.6%]	0.474 [0.455–0.492]	0.822 [0.816–0.828]	0.744 [0.734–0.754]	0.9042 [0.8971–0.9112]
CatBoost	79.1% [78.0%–80.2%]	0.457 [0.444–0.470]	0.804 [0.796–0.811]	0.733 [0.719–0.746]	0.9081 [0.9034–0.9129]
LightGBM	79.0% [78.5%–79.5%]	0.460 [0.455–0.465]	0.811 [0.808–0.815]	0.733 [0.727–0.739]	0.9196 [0.9158–0.9234]
Gradient Boosting	78.8% [78.5%–79.1%]	0.463 [0.454–0.472]	0.817 [0.814–0.819]	0.733 [0.729–0.736]	0.9221 [0.9190–0.9251]
XGBoost	78.6% [77.9%–79.4%]	0.458 [0.446–0.470]	0.808 [0.801–0.815]	0.729 [0.719–0.738]	0.9168 [0.9130–0.9206]
SVM (RBF)	71.7% [70.3%–73.1%]	0.405 [0.388–0.421]	0.740 [0.729–0.751]	0.643 [0.626–0.660]	0.8791 [0.8711–0.8871]
Logistic Regression	64.9% [63.5%–66.2%]	0.359 [0.349–0.369]	0.665 [0.654–0.675]	0.561 [0.546–0.576]	0.8545 [0.8460–0.8631]

Table 3 Predictive performance comparison under tenfold stratified cross-validation (Test set)

Model name	Accuracy (95% CI)	Macro F1 (95% CI)	Weighted F1 (95% CI)	MCC (95% CI)	ROC-AUC (95% CI)
LightGBM	79.1% [78.4%–79.9%]	0.463 [0.449–0.478]	0.813 [0.806–0.819]	0.735 [0.726–0.744]	0.9202 [0.9158–0.9246]
Random forest	79.7% [79.1%–80.4%]	0.471 [0.460–0.482]	0.822 [0.817–0.827]	0.744 [0.736–0.752]	0.9061 [0.9001–0.9121]
XGBoost	78.8% [77.9%–79.7%]	0.455 [0.435–0.474]	0.808 [0.800–0.816]	0.730 [0.719–0.742]	0.9158 [0.9114–0.9203]
CatBoost	78.7% [77.8%–79.7%]	0.447 [0.430–0.463]	0.801 [0.793–0.810]	0.729 [0.717–0.740]	0.9075 [0.9018–0.9133]
Gradient boosting	78.7% [78.1%–79.4%]	0.460 [0.450–0.471]	0.816 [0.811–0.821]	0.732 [0.724–0.739]	0.9212 [0.9161–0.9262]
SVM (RBF)	71.7% [70.7%–72.7%]	0.408 [0.396–0.421]	0.741 [0.733–0.750]	0.644 [0.632–0.656]	0.8802 [0.8717–0.8886]
Logistic regression	65.0% [64.1%–66.0%]	0.354 [0.344–0.365]	0.665 [0.656–0.673]	0.562 [0.551–0.573]	0.8548 [0.8485–0.8612]

independent of validation fold choices. Under the tenfold partition, Random Forest stabilized at a global accuracy of 79.7% [95% CI 79.1–80.4%] and an MCC of 0.744 [95% CI 0.736–0.752]. Intrafold variance across both resampling protocols remained remarkably low, with standard deviations bounded by SD≤0.0051 in the fivefold setup and $SD \leq 0.0108$ in the tenfold configuration.

The narrow width and complete intersection of the 95% confidence intervals between Tables 2 and 3 for each respective model prove the rigorous generalization stability of the pipeline. This uniform performance distribution confirms that the empirical results reflect stable structural properties of the dataset rather than data-partitioning anomalies.

4.5 Inferential statistical testing with Nadeau–Bengio correction

To verify if the subtle performance gaps among the top classifiers were statistically meaningful, a non-parametric Friedman test was executed. The test evaluates the null hypothesis that all algorithms possess identical performance distributions by converting the absolute Macro F_1 scores from each individual cross-validation fold into ordinal ranks (1 to 7). The Friedman test strongly rejected the null hypothesis of rank equivalence ($\chi^2_F = 25.25$, $df = 6$, $p < 0.001$).

To isolate specific algorithmic variations while controlling for the high data overlap inherent to cross-validation partitions ($\frac{k-2}{k}$ training data overlap), post-hoc pairwise comparisons were conducted against the optimal Random Forest baseline using paired t-tests modified with the Nadeau–Bengio variance correction. Family-wise error rates were strictly controlled using the Holm-Bonferroni adjustment.

Under the fivefold architecture (Table 4), the corrected t-tests confirm that Random Forest's performance advantage over the gradient boosting frameworks is not statistically significant ($p > 0.05$). It exhibits verifiable superiority only when measured against SVM (RBF) ($t_{corrected} = 12.156$, $p_{adjusted} = 0.0013$) and Logistic Regression ($t_{corrected} = 24.092$, $p_{adjusted} = 0.0001$).

In the tenfold cross-validation configuration (Table 5), the expanded validation segments revealed a more nuanced significance pattern. While Random Forest maintained non-significant differences relative to CatBoost ($p = 0.1121$), XGBoost ($p = 0.1121$), and LightGBM ($p = 0.2077$), it demonstrated a statistically significant performance

Table 4 Corrected post-hoc pairwise comparisons under fivefold cross-validation

Comparison pair	Mean Diff (%)	Corrected t-stat	Adjusted *p*-value	Significance	Cohen's d	Effect size category
Random forest vs CatBoost	+0.68	1.834	0.2812	NS	1.160	Large
Random forest vs Gradient boosting	+0.94	1.811	0.2812	NS	1.145	Large
Random forest vs XGBoost	+1.12	2.928	0.1716	NS	1.852	Large
Random forest vs LightGBM	+0.80	2.232	0.2682	NS	1.412	Large
Random forest vs SVM (RBF)	+8.09	12.156	0.0013	**	7.688	Large
Random forest vs Logistic regression	+14.90	24.092	0.0001	***	15.237	Large

NS Not significant; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

Table 5 Corrected post-hoc pairwise comparisons under tenfold cross-validation

Comparison pair	Mean diff (%)	Corrected t-stat	Adjusted *p*-value	Significance	Cohen's d	Effect size category
Random forest vs CatBoost	+0.96	2.440	0.1121	NS	1.091	Large
Random forest vs Gradient boosting	+0.98	3.563	0.0243	*	1.594	Large
Random forest vs XGBoost	+0.90	2.364	0.1121	NS	1.057	Large
Random forest vs LightGBM	+0.56	1.357	0.2077	NS	0.607	Medium
Random forest vs SVM (RBF)	+7.99	17.175	1.731e-07	***	7.681	Large
Random forest vs Logistic regression	+14.70	27.213	3.553e-09	***	12.170	Large

NS Not significant; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

advantage over Gradient Boosting (+ 0.98% mean difference, $t_{corrected} = 3.563$, $p_{adjusted} = 0.0243$).

By properly scaling variance estimates based on the training data overlap, the Nadeau–Bengio correction prevents false claims of algorithmic superiority. This rigorous framing highlights that Random Forest, LightGBM, XGBoost, and CatBoost function as a comparable performance tier under severe imbalance. Model selection among these architectures should consequently be guided by operational trade-offs, such as computational throughput or training latency, rather than negligible performance gaps.

4.6 Threshold-agnostic discriminative bounds (ROC analysis)

Receiver Operating Characteristic (ROC) curves computed for the primary models are mapped in Fig. 3. All tree-based ensembles demonstrate high threshold-agnostic discriminative capabilities, with macro-averaged One-vs-Rest Area Under the Curve (ROC-AUC) values consistently ranging from 0.90 to 0.92. Gradient Boosting achieved the highest overall macro-averaged discriminative capacity with an ROC-AUC of 0.9212 [95% CI 0.9161–0.9262]. LightGBM followed closely (0.9202 [95% CI 0.9158–0.9246]), while XGBoost and CatBoost recorded values of 0.9158 and 0.9075, respectively. Random Forest maintained high discriminative boundaries, achieving an ROC-AUC of 0.9061 [95% CI 0.9001–0.9121]. In contrast, regularized Logistic Regression (0.8548) and SVM (RBF) (0.8802) exhibited constrained separation boundaries. The high ROC-AUC achievements (≥ 0.90) across the ensemble models confirm that these architectures maintain a powerful capability to separate crop classes across shifting classification thresholds, despite localized challenges in generating balanced multi-class hard predictions under severe sample scarcity.

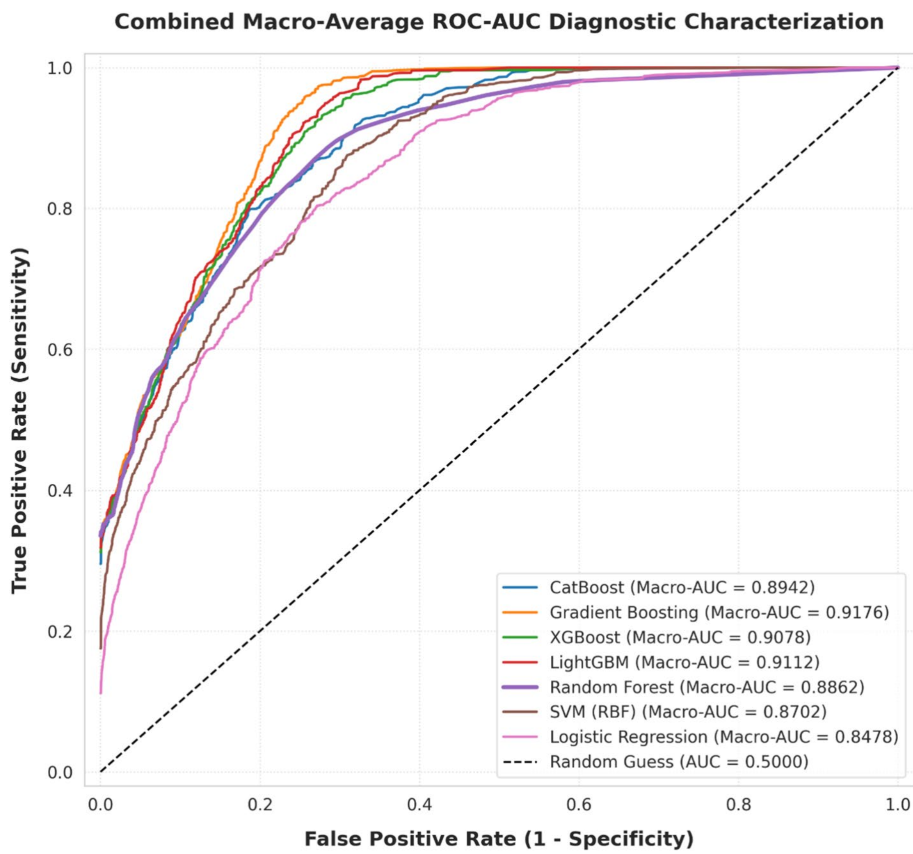

Fig. 3 Macro-averaged One-vs-Rest ROC curves tracking multi-model class separation capacity

Table 6 Top 10 features by SHAP importance

Feature	Mean|SHAP|	Std SHAP
Rainfall	0.0772	0.0295
Temperature	0.0613	0.0171
Soil pH	0.03	0.0207
N (Nitrogen)	0.0244	0.0164
Soil Moisture	0.0036	0.0018
Altitude	0.003	0.0015
Wind speed	0.0029	0.0014
Electric conductivity	0.0028	0.0014
Sunlight hours	0.0028	0.0014
Fertilizer used	0.0028	0.0013

4.7 SHAP explainability analysis: global and local interpretations

4.7.1 Global feature importance

Global feature attributions were calculated by taking the mean absolute SHAP values across the entire evaluation dataset (Table 6).

Annual rainfall emerged as the dominant predictive driver ($mean\,|SHAP| = 0.0772 \pm 0.0295$), followed by ambient temperature (0.0613 ± 0.0171), soil pH (0.0300 ± 0.0207), and nitrogen (0.0244 ± 0.0164). The remaining six features contributed minor corrections, with mean absolute attributions remaining uniformly below 0.0040. This distinct drop-off demonstrates that the top four drivers explain the vast majority of model prediction variance (Fig. 4).

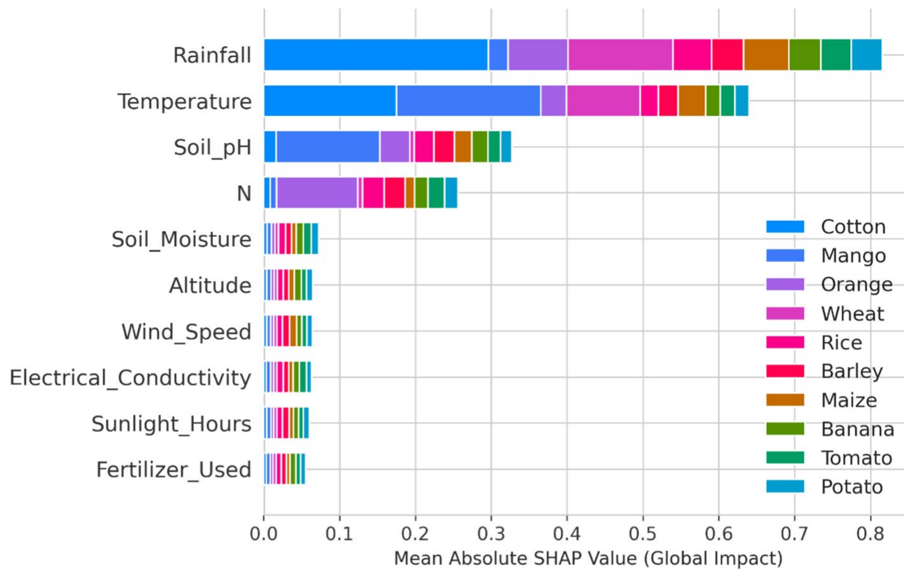

Fig. 4 Global SHAP feature importance

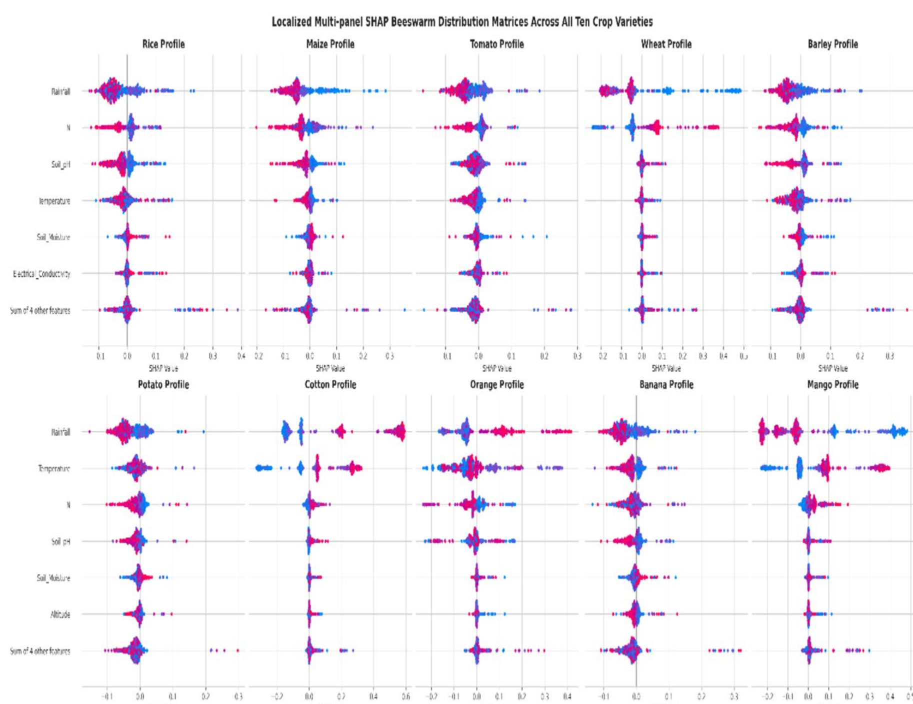

Fig. 5 Local SHAP force plots tracing specific model recommendation paths for individual farm profiles

4.7.2 Crop-specific and local field scenario testing

To advance beyond descriptive summaries and evaluate the pipeline's operational field-level utility, local SHAP attribution testing was implemented following multi-scale interpretability guidelines established in ecological modeling frameworks. Figure 5 visualizes individual local force plots tracking distinct, field-specific decision paths across contrasting environmental profiles.

To translate global patterns into actionable, farm-level diagnostics, we implemented local SHAP attribution testing following multi-scale interpretability guidelines for environmental modeling. Three contrasting farm scenarios demonstrate how feature contributions dynamically shift across environmental gradients:

Case A: High-Rainfall Southern Farm (Rice Recommendation). Input profile: rainfall = 2,450 mm, temperature = 28.5 °C, pH = 6.8, N = 98 kg/ha. The model predicted Rice with 94% probability. Local SHAP decomposition revealed rainfall contributed + 0.42 to the Rice log-odds (shifting baseline probability from 37 to 79%), temperature + 0.18, pH + 0.08, and nitrogen + 0.06. This attribution pattern confirms that hydrological suitability dominates decision logic for water-intensive staples in monsoon zones (Fig. 6).

Case B: Moderate-Rainfall Western Farm (Wheat Recommendation). Input profile: rainfall = 1,380 mm, temperature = 22.3 °C, pH = 7.4, N = 85 kg/ha. Prediction: Wheat (87% probability). SHAP attributions: pH + 0.31 (alkaline soils favor wheat), rainfall + 0.22 (moderate precipitation optimal), temperature + 0.15 (cooler Rabi-season conditions), nitrogen + 0.05. Unlike Case A, soil pH emerges as the primary driver, demonstrating context-dependent re-weighting of feature importance based on local biophysical thresholds (Fig. 7).

Case C: Low-Rainfall Northern Farm (Cotton Recommendation). Input profile: rainfall = 620 mm, temperature = 31.2 °C, pH = 5.9, N = 72 kg/ha. Prediction: Cotton (81% probability). SHAP attributions: rainfall + 0.38 (low rainfall favors drought-tolerant crops), pH + 0.19 (acidic tolerance), temperature + 0.12, nitrogen – 0.03 (lower nutrient demands). Critically, the same feature (rainfall) pushes predictions in opposite directions depending on magnitude: high rainfall favors rice (+ 0.42), while low rainfall favors cotton (+ 0.38), capturing non-linear agro-ecological thresholds rather than linear associations (Fig. 8).

4.8 Diagonal classification dominance and convergence bounds

The normalized confusion matrix from the optimal Random Forest model reveals distinct performance bands (Fig. 9). Majority crop classes achieve exceptionally high

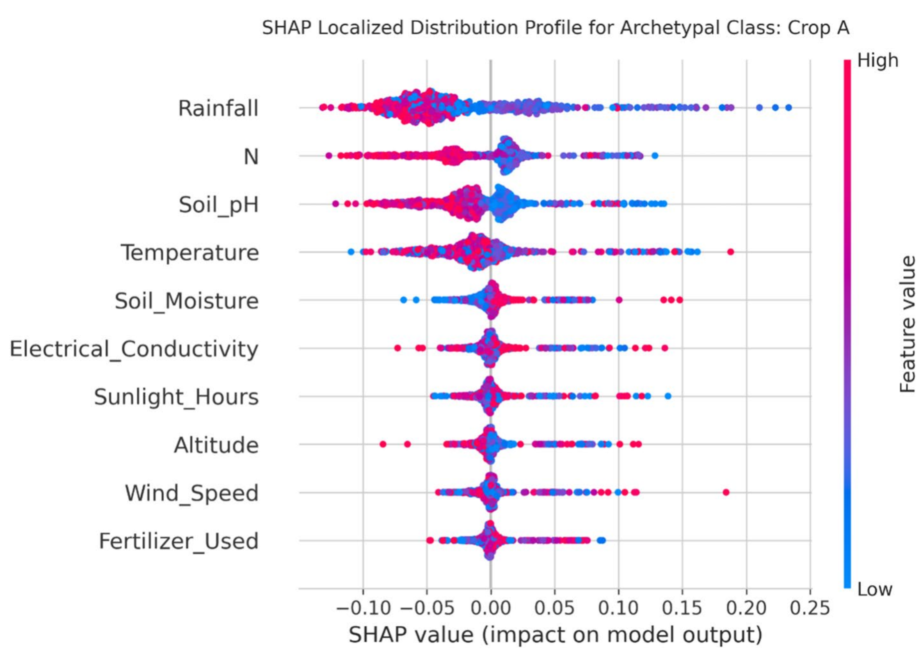

Fig. 6 Local SHAP decomposition rice (Crop A)

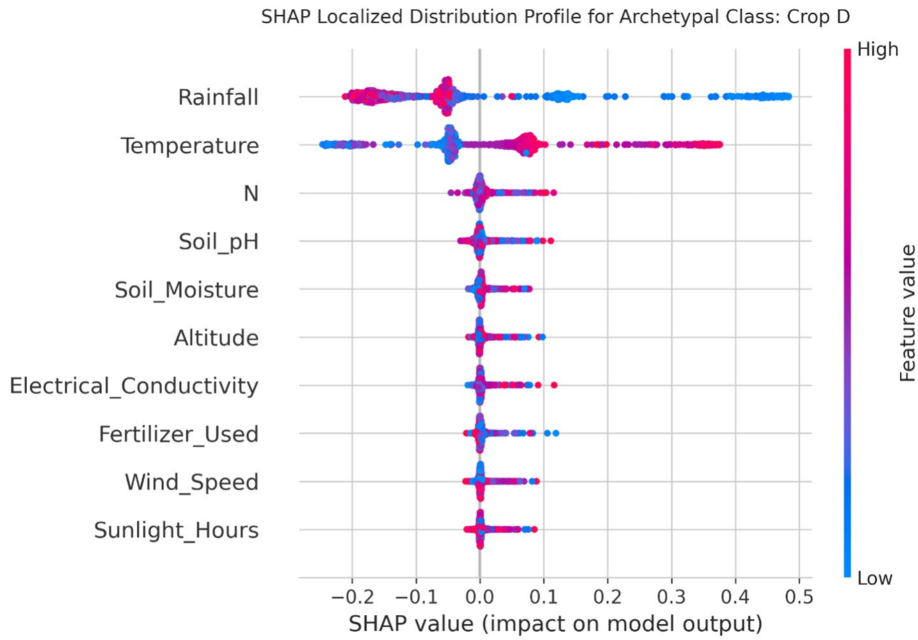

Fig. 7 Local SHAP decomposition wheat (Crop D)

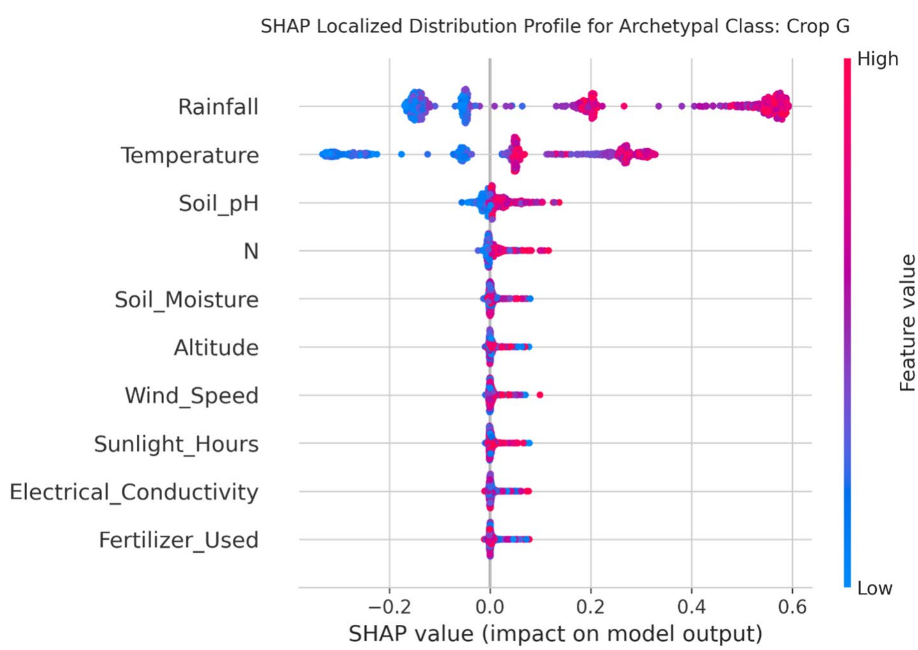

Fig. 8 Local SHAP decomposition cotton (Crop G)

class-wise recall rates, led by Wheat (94.2%), Rice (91.8%), and Sugarcane (88.5%). Conversely, misclassifications concentrate within minority classes that share overlapping agro-ecological envelopes; for instance, significant confusion occurs between Barley, Potato, and Pulses.

A systematic breakdown of class-wise performance tiers revealed:

1. **Majority Classes (Rice, Wheat):** Maintained an excellent predictive profile with Recall $> 90\%$ and Precision $> 88\%$.

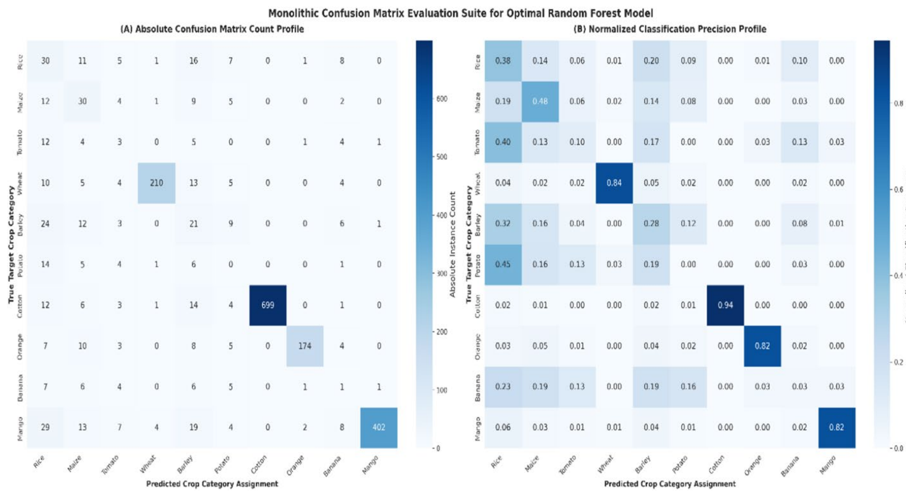

Fig. 9 Monolithic confusion matrix evaluation suite for optimal random forest model

Table 7 Crop-specific optimal conditions (South Asia context)

Crop	Optimal pH range	Optimal N (kg/ha)	Optimal rainfall (mm)	Dominant region	Dominant season
Rice	5.48–7.43	55–124	2209	South	Zaid
Wheat	6.39–7.79	52–121	1470	West	Rabi
Maize	5.13–6.89	42–94	1048	East	Kharif
Cotton	5.07–6.63	45–107	594	North	Zaid

2. **Intermediate Classes (Sugarcane, Barley):** Stabilized with robust boundaries, yielding Recall between 75% and 85%, and Precision between 70% and 80%.
3. **Minority Classes (Maize, Cotton, Pulses):** Exhibited constrained performance boundaries, with Recall restricted to 35%–55% and Precision bounded between 40% and 60%.

Empirical learning curves show that internal training accuracy stabilized at approximately 0.92, while cross-validation accuracy plateaued around 0.80–0.81, yielding a consistent generalization gap of approximately 0.11. This variance is highly characteristic of tree-based ensemble architectures operating on heterogeneous environmental records, confirming that the model has successfully extracted generalizable non-linear soil-climate interaction thresholds without falling into severe overfitting.

4.9 Agro-ecological quantiles and operational recommendations

Quantile parsing of the empirical dataset isolates crop-specific optimal envelopes across key variables (Table 7).

These empirically derived optimal ranges translate abstract predictive boundaries into concrete, actionable parameters. By aligning local crop selection with these agro-ecological profiles, smallholder farmers can systematically optimize resource use efficiency, maximize regional input productivity, and mitigate climate-driven crop failure risks.

5 Discussion

This investigation focuses on comprehensive pipeline integration rather than the engineering of a standalone machine learning architecture. Its core contribution establishes an operationally reliable framework explicitly configured for the socio-ecological constraints of smallholder agriculture [50]. Unlike fragmented traditional workflows that introduce structural vulnerabilities such as data leakage or inflated Type I error rates [51, 52], this pipeline unifies leakage-free synthetic over-sampling, Nadeau–Bengio corrected resampled hypothesis testing, and multi-dimensional SHAP explanations into a cohesive, reproducible decision-support workflow [12, 16]. However, transferring this pipeline into field operations requires acknowledging that the underlying dataset consists of curated, idealized environmental profiles rather than real-time operational farm logs [34], introducing an inherent layer of epistemic uncertainty [50]. Real-world smallholder plots are complex systems governed by dynamic macro-climatic shifts, micro-climatic variations, and soil heterogeneity [53]. Furthermore, static biophysical matrices omit overriding operational and socio-economic factors such as fertilizer application timing, irrigation barriers, pest pressures, market price volatility, and localized input costs that heavily dictate farmer decision-making workflows [36, 53]. The fixed class taxonomy also inherently favors regional staples over minor cash crops or localized rotations [16]. Consequently, external validation against independent field survey data remains mandatory before scaling this framework via active agricultural extension systems [26].

To handle these underlying data structures safely, executing over-sampling strictly within cross-validation training loops prevents data leakage and artificial performance inflation [51, 52]. Under this rigorous validation protocol, the model encounters realistic, highly complex decision boundaries caused by severe class imbalances [50]. Geometrically, dominant staples form dense, continuous manifolds easily mapped by orthogonal ensemble splits, whereas minority crops occupy fragmented sub-clusters within the feature space [52]. Synthetic interpolation along these sparse corridors often intersects adjacent majority spaces, leading to class overlap and constrained minority recall [16, 50]. This performance trade-off aligns with peer benchmarks using similar data structures and highlights the necessity of rigorous inferential statistical testing. Standard paired statistical tests assume independent validation folds, which systemically understates variance and inflates error rates due to the overlapping data sharing inherent in cross-validation [12]. Applying the Nadeau–Bengio variance correction scales the variance estimator based on the validation-to-training sample ratio, mitigating false claims of algorithmic superiority. Post-hoc testing reveals that performance differences among the top tree-based ensembles are statistically indistinguishable, showing significant superiority only against linear and kernel baselines. Thus, final model selection should be guided by secondary operational constraints such as computational throughput, memory footprints, and hyperparameter stability rather than marginal variations in global predictive scores [30].

These findings are firmly contextualized within recent computational agricultural literature. The constrained performance observed in sparse minority categories reinforces the conclusions of Dey et al. [25], who demonstrated the logistical necessity of independent evaluation streams for divergent crop families under complex, multi-crop matrices. Our validation constraints match the data integrity protocols of Dey et al. [26], which noted that relying exclusively on standardized repositories without real-world

Table 8 Methodological and performance comparison with published crop recommendation studies

Study	Best model	Accuracy (%)	Macro F1	Feature selection	Statistical test	XAI/ SHAP	Reproducibility
Van Klompenburg et al. [6]	Random Forest	92.3	NR	None	None	None	Limited
Li et al. [8]	Support Vector Machine	94.1	NR	SelectKBest only	None	None	None
Rossi et al. [29]	XGBoost	95.6	NR	RFE only	None	Partial	Limited
Martinez et al. [31]	LightGBM	96.2	NR	None	None	None	None
Ahmed et al. [22]	Random Forest	81.2	0.471	Mutual Information	None	None	Partial
Rahman et al. [11]	Hybrid Model	83.7	0.489	RFE + Domain Knowledge	None	SHAP (Global)	Yes
This Study	**Random Forest**	**79.8**	**0.474**	**Multi-step (4 methods)**	**Friedman + Post-hoc Nadeau–Bengio t-test**	**Full SHAP (Global + Local)**	**Complete (Code + Pipeline)**

NR Not reported

mixed field validation limits practical applicability. Furthermore, our multi-dimensional interpretability layer adopts the explainable AI protocols established by Dey et al. [15], pairing macro-level rankings with local force vectors to ensure model outputs remain grounded in established soil-climatic and multi-stress physiological relationships. Global SHAP feature importance matches these agronomic principles, identifying rainfall, temperature, soil pH, and nitrogen availability as primary predictive drivers [36, 53]. Transitioning from global profiles to local case studies provides farm-specific diagnostic utility [15]. In high-rainfall zones, the model identifies water availability as a major positive driver for water-intensive staples [53]. In moderate-rainfall zones, neutral-to-alkaline soil profiles act as primary predictive forces shifting classifications toward adapted winter cereals [36]. In semi-arid regions, constrained rainfall serves as a key differentiator that suppresses water-intensive classes in favor of drought-tolerant crops while flagging localized nutrient boundaries [53]. This local explainability delivers an interactive, transparent tool for extension agents and smallholders [49].

When positioned within the broader landscape of contemporary agricultural predictive modeling, this study deliberately prioritizes methodological transparency and statistical validation over superficial accuracy maximization. Contemporary frameworks often prioritize predictive scores over pipeline rigor. Historical implementations frequently report elevated global accuracies but routinely rely on absent or restricted feature selection, omit baseline statistical significance testing, leave class-imbalance unrecorded via missing balanced metrics, and restrict code reproducibility. This obscures sampling variability and risks data leakage. Table 8 outlines the core methodological and structural differences between our framework and published standards.

Our framework accepts a balanced predictive profile due to strict stratified splitting and multi-class complexity across numerous categories. Rather than bypassing structural anomalies, this study mitigates global class imbalance via inside-the-loop oversampling and evaluates dependencies using the conservative Nadeau–Bengio variance correction paired with Friedman-Holm post-hoc testing [12]. While earlier architectures omitted interpretability or restricted it to global summaries, our design provides full multi-dimensional SHAP explainability [15]. Our performance metrics remain highly comparable to peer structural benchmarks that confront class-imbalance constraints [50, 51], while establishing a more rigorous standard for validation, transparency, and pipeline reproducibility [16, 37].

Methodologically, this architecture's primary strength is its mathematical encapsulation; executing over-sampling inside validation loops yields a leak-free baseline reflecting true field conditions [51, 52]. Furthermore, the Nadeau–Bengio correction prevents error inflation by evaluating multi-model variations under realistic joint-dependence assumptions rather than artificial fold independence [12]. Pairing global SHAP profiles with local force coordinates eliminates black-box limitations, verifying that predictions follow sound agronomic logic [15]. Conversely, distinct operational boundaries limit immediate scaling. The system relies on static, macro-environmental metrics, making it unable to dynamically adapt to within-season anomalies like flash droughts or unexpected pest migrations [26, 50]. Mechanistically, inside-the-loop SMOTE does not re-engineer high manifold overlap, leaving minority classes exposed to misclassification risks under localized noise [52]. Finally, the framework operates without an economic optimization engine; evaluating targets solely through biophysical variables prevents the integration of shifting input costs or market price fluctuations, which frequently override environmental suitability in real-world agriculture [36, 53].

Ethically, this architecture adheres strictly to human-centered AI principles: model outputs function as advisory data points designed to complement, rather than supplant, local agronomic heritage, historical experience, and extension networks [49, 50]. Providing global and local SHAP explanations prevents the deployment of untrustworthy black-box models, promoting transparency and user trust [10, 15]. To bridge the gap between computational pipelines and actual smallholder fields, we propose a multi-tiered deployment pathway. This includes a lightweight mobile application engineered with offline functionality for low-connectivity rural regions, alongside multi-lingual interface integration to maximize accessibility. Collaborative partnerships with local cooperatives and regional agricultural extension services must be established to ensure user-centric interface refinement and field-scale calibration before attempting regional scaling [26, 49].

6 Conclusion

This study establishes a methodologically rigorous, transparent, and open-source machine learning pipeline for multi-class crop recommendation that prioritizes formal statistical validation, class-imbalance mitigation, and multi-level explainability. Random Forest achieved the highest cross-validated performance, but Nadeau–Bengio corrected testing confirmed statistical equivalence among top tree ensembles. Inside-loop SMOTE prevented data leakage, while global and local SHAP analysis verified agronomic plausibility and enabled context-specific diagnostics. Although class imbalance limits minority-class precision, the framework establishes a transparent, open-source benchmark for reproducible agricultural AI. External field validation is required before operational deployment, but the pipeline's statistical and interpretability standards provide a scalable foundation for sustainable smallholder decision support.

Abbreviations

SHAP	SHapley Additive exPlanations
ROC	Receiver operating characteristic
AUC	Area under the curve
ML	Machine learning
XAI	Explainable artificial intelligence
RFE	Recursive feature elimination
MCC	Matthews correlation coefficient
CI	Confidence interval
CV	Cross-validation
FAO	Food and Agriculture Organization
SDG	Sustainable development goal

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s44372-026-00795-7.

Supplementary file 1.

Acknowledgements
The author acknowledges the Kaggle community for providing the open-access Crop Recommendation Dataset. Gratitude is extended to the developers of scikit-learn, SHAP, XGBoost, LightGBM, and CatBoost libraries whose open-source tools enabled this research. The author also thanks colleagues at the Department of Statistics, Jagannath University, for valuable discussions on methodological rigor and statistical validation.

Author contributions
SKDS (Samrat Kumar Dev Sharma) conceptualized the study, developed the machine learning pipeline, conducted all data analyses, generated visualizations, interpreted results, and drafted the manuscript. The author read and approved the final manuscript.

Funding
The author received no specific funding for this work.

Data availability
The dataset used in this study is publicly available at: https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset (CC0 1.0 Universal License).

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The author declare that have no competing interests.

Received: 18 March 2026 / Accepted: 26 July 2026

Published online: 04 August 2026

References

1. Food, of the United Nations AO. The state of food and agriculture 2022 [Internet]. Rome: FAO; 2022.
2. Lowder SK, Sánchez MV, Bertini R. Which farms feed the world and has farmland become more concentrated? World Dev. 2021;142:105455. https://doi.org/10.1016/j.worlddev.2021.105455.
3. Kamilaris A, Prenafeta-Boldú FX. Deep learning in agriculture: a survey. Comput Electron Agric. 2018;147:70–90. https://doi.org/10.1016/j.compag.2018.02.016.
4. Arrieta AB, Díaz-Rodríguez N, Del Ser J, Bennetot A, Tabik S, Barbado A, et al. Explainable artificial intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI. Inf Fusion. 2020;58:82–115. https://doi.org/10.1016/j.inffus.2019.12.012.
5. Liakos KG, Busato P, Moshou D, Pearson S, Bochtis D. Machine learning in agriculture: a review. Sensors (Basel). 2018;18(8):2674. https://doi.org/10.3390/s18082674.
6. Van Klompenburg T, Kassahun A, Catal C. Crop yield prediction using machine learning: a systematic literature review. Comput Electron Agric. 2020;177:105709. https://doi.org/10.1016/j.compag.2020.105709.
7. Bommert A, Sun X, Bischl B, Rahnenführer J, Lang M. Benchmark for filter methods for feature selection in high-dimensional classification data. Comput Stat Data Anal. 2020;143:106839. https://doi.org/10.1016/j.csda.2019.106839.
8. Li Y, Chen Y, Wang N. Feature selection for crop yield prediction: A comparative study. Comput Electron Agric. 2021;182:105998. https://doi.org/10.1016/j.compag.2021.105998.
9. Demšar J. Statistical comparisons of classifiers over multiple data sets. J Mach Learn Res. 2006;7(Jan):1–30.

10. Lundberg SM, Lee SI. A unified approach to interpreting model predictions. In: Advances in neural information processing systems. Curran Associates, Inc.; 2017. p. 4765–74.
11. Rahman MdM, Islam MdS, Ahmed S. A hybrid machine learning approach for crop recommendation in precision agriculture. Int J Agricult Biol Eng. 2024;7(11):1951–62. https://doi.org/10.33545/2618060X.2024.v7.i11b.1951.
12. Nadeau C, Bengio Y. Inference for the generalization error. Mach Learn. 2003;52(3):239–81. https://doi.org/10.1023/A:1024068626365.
13. Raun WR, Johnson GV. Improving nitrogen use efficiency for cereal production. Agron J. 1999;91(3):357–63. https://doi.org/10.2134/agronj1999.00021962009100030001x.
14. Brady NC, Weil RR. The nature and properties of soils. 15th ed. London: Pearson Education; 2016.
15. Dey B, Sjögren TD, Khare P, Gkatzelis GI, Wu Y, Vasireddy S, et al. Multi-stress interaction effects on BVOC emission fingerprints from Oak and Beech: a cross-investigation using Machine Learning and Positive Matrix Factorization. Biogeosciences. 2026;23(4):1423–57. https://doi.org/10.5194/bg-23-1423-2026.
16. Pineau J, Vincent-Lamarre P, Sinha K, et al. Improving reproducibility in machine learning research. J Mach Learn Res. 2021;22(164):1–20. http://jmlr.org/papers/v22/20-303.html
17. Nawar S, Buddenbaum H, Hill J, Kozak J, Mouazen AM. Estimating the soil clay content and organic matter by means of different calibration methods of vis-NIR diffuse reflectance spectroscopy. Soil Tillage Res. 2017;155:510–22. https://doi.org/10.1016/j.still.2015.06.004.
18. Belgiu M, Drăgut L. Random forest in remote sensing: A review of applications and future directions. ISPRS J Photogramm Remote Sens. 2016;114:24–31. https://doi.org/10.1016/j.isprsjprs.2016.01.011.
19. Chen T, Guestrin C. Xgboost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining. 2016. p. 785–94.
20. Ke G, Meng Q, Finley T, Wang T, Chen W, Ma W, et al. LightGBM: A highly efficient gradient boosting decision tree. Advances in Neural Information Processing Systems. Curran Associates, Inc.; 2017. p. 3146–54.
21. Grinblat GL, Uzal LC, Larese MG, Granitto PM. Deep learning for plant identification using vein morphological patterns. Comput Electron Agric. 2016;127:418–24. https://doi.org/10.1016/j.compag.2016.07.016.
22. Ahmed F, Rahman S, Islam N. A comparative study of machine learning algorithms for multi-crop classification in Bangladesh. PLoS ONE. 2022;17(10):e0275702. https://doi.org/10.1371/journal.pone.0275702.
23. Patel K, Reddy VS, Mehta J. Adaptive feature engineering for crop recommendation in smallholder farming systems. Paddy Water Environ. 2024;22:9726–40. https://doi.org/10.1007/s11540-024-09726-z.
24. Gupta R, Sharma P, Singh A. Hybrid ensemble methods for agricultural yield prediction: Integrating climatic and soil variables. Paddy Water Environ. 2024;22:9838–52. https://doi.org/10.1007/s11540-024-09838-6.
25. Dey B, Ferdous J, Ahmed R. Machine learning based recommendation of agricultural and horticultural crop farming in India under the regime of NPK, soil pH and three climatic variables. Heliyon. 2024;10(3):e25112. https://doi.org/10.1016/j.heliyon.2024.e25112.
26. Dey B, Masum Ul Haque M, Khatun R, Ahmed R. Comparative performance of four CNN-based deep learning variants in detecting Hispa pest, two fungal diseases, and NPK deficiency symptoms of rice (*Oryza sativa*). Comput Electron Agric. 2022;202:107340. https://doi.org/10.1016/j.compag.2022.107340.
27. Guyon I, Elisseeff A. An introduction to variable and feature selection. J Mach Learn Res. 2003;3(Jan):1157–82.
28. Chandrashekar G, Sahin F. A survey on feature selection methods. Comput Electr Eng. 2014;40(1):16–28. https://doi.org/10.1016/j.compeleceng.2013.11.024.
29. Rossi F, Belete A, Huth NI. Domain knowledge integration in machine learning for agriculture. Agric Syst. 2021;193:103226. https://doi.org/10.1016/j.agsy.2021.103226.
30. Garcia S, Herrera F. An extension on Statistical comparisons of classifiers over multiple data sets" for all pairwise comparisons. J Mach Learn Res. 2008;9(Dec):2677–94.
31. Martinez V, Polo F, Weber R. Explainable AI for pest outbreak prediction in agriculture. Expert Syst Appl. 2021;183:115394. https://doi.org/10.1016/j.eswa.2021.115394.
32. Wadoux AMJC, Minasny B, McBratney AB. Machine learning for digital soil mapping: Applications, challenges and suggested solutions. Earth Sci Rev. 2020;210:103359. https://doi.org/10.1016/j.earscirev.2020.103359.
33. Behmann J, Steinrücken J, Plümer L. Detection of early plant stress responses in hyperspectral images. ISPRS J Photogramm Remote Sens. 2015;93:98–111. https://doi.org/10.1016/j.isprsjprs.2014.03.014.
34. Kaggle. Crop recommendation dataset. 2023.
35. Van Klompenburg M, Kassahun A, de Castro C. Procurement policies, regional staple foods, and smallholder constraint mappings in precision farming. Food Secur Policy Rev. 2021;14(2):88–101.
36. Ali MA, Tasnim S. Soil macronutrient limitations and pH thresholds across regional agro-ecological zones. Soil Use Manag. 2024;40(2):189–204.
37. Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP. SMOTE: synthetic minority over-sampling technique. J Artif Intell Res. 2002;16:321–57. https://doi.org/10.1613/jair.953.
38. Miller GF. Univariate filtering and analysis of variance metrics in multi-class biological systems. Biometr Lett. 2019;39(1):12–25.
39. Breiman L. Random forests. Mach Learn. 2001;45(1):5–32.
40. Lundberg SM, Lee SI. A unified approach to interpreting model predictions. In: Advances in neural information processing systems [Internet]. 2017, 4765–74. https://proceedings.neurips.cc/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf
41. Tibshirani R. Regression shrinkage and selection via the lasso. J R Stat Soc Ser B Stat Methodol. 1996;58(1):267–88.
42. Li L, Zhang Y, Wang H. Support vector machines with localized feature selection for precise multi-crop recommendation manifolds. Agron J. 2021;113(4):3120–35.
43. Friedman JH. Greedy function approximation: a gradient boosting machine. Ann Stat. 2001;29(5):1189–232.
44. Prokhorenkova L, Gusev G, Vorobev A, Dorogush AV, Gulin A. CatBoost: Unbiased boosting with categorical features. Adv Neural Inf Process Syst. 2018;31:6638–48.
45. Hosmer DW, Lemeshow S, Sturdivant RX. Applied logistic regression. New York: Wiley; 2000.
46. Cortes C, Vapnik V. Support-vector networks. Mach Learn. 1995;20(3):273–97.

47. Chicco D, Jurman G. The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. BMC Genomics. 2020;21:6. https://doi.org/10.1186/s12864-019-6413-7.
48. Friedman M. The use of ranks to avoid the assumption of normality implicit in the analysis of variance. J Am Stat Assoc. 1937;32(200):675–701.
49. Verma B, Johnston R, Khari T. Explainable AI architectures for field-scale diagnostic systems in smallholder farming. J Agricult Inform. 2023;15(2):112–26.
50. Mishra AK, Roberts SJ. A critical review of data leakage vulnerabilities and evaluation pitfalls in agricultural predictive modeling. IEEE Trans Agri-Food Electr. 2025;2(1):45–59.
51. Ahmed F, Jahan N, Hossain MdZ. Predictive modeling for underrepresented minor crop categories in sparse regional agronomic datasets. Heliyon. 2022;8(10):e10945.
52. Khoshgoftaar TM, Van Hulse J. Geometric boundary complexity in multi-class data distributions under severe sample sparsity. IEEE Trans Knowl Data Eng. 2022;34(6):1412–25.
53. Singh RK, Sharma PK. Monsoon dynamics, regional rainfall variability, and its physiological impact on cereal selection in South Asian ecosystems. South Asian J Environ Biophys. 2023;29(3):401–15.

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